Socioeconomic Inequalities in Adolescent Childbearing in Malawi

Analysis of the 2017 Malawi Demographic and Health Survey

Research question

Does adolescent childbearing differ across household wealth groups in Malawi?

Background

Adolescent childbearing remains an important public health concern in Malawi because of its potential health, educational and socioeconomic consequences for young girls. Early childbearing may expose adolescent mothers and their infants to health risks, while also affecting girls' educational opportunities and future economic prospects. Adolescent mothers may experience difficulties continuing their education and participating in economic activities, potentially reinforcing existing socioeconomic disadvantages.

Household socioeconomic circumstances are important in understanding differences in adolescent childbearing. Household wealth may be associated with access to education, healthcare services and reproductive health information, as well as the opportunities available to adolescent girls. Girls from poorer households may experience additional barriers, including limited access to reproductive health services and greater socioeconomic pressures that influence the timing of childbearing. Consequently, adolescent childbearing may be unevenly distributed across socioeconomic groups.

Understanding how adolescent childbearing varies across household wealth groups can help identify inequalities and inform public health discussions about equitable access to adolescent sexual and reproductive health services. This study therefore examines the prevalence of adolescent childbearing among girls aged 15–19 years across household wealth quintiles using data from the 2017 Malawi Demographic and Health Survey.

Methods

Data source and study population

This study used data from the 2017 Malawi Demographic and Health Survey (DHS), obtained through the Integrated Public Use Microdata Series (IPUMS) DHS database. The analysis was restricted to girls aged 15–19 years. The final analytical sample consisted of 828 girls.

Study variables

The outcome variable was adolescent childbearing, defined as having ever given birth to at least one child. It was coded as a binary variable, distinguishing girls who had ever had a child from those who had not. Household wealth quintile was the explanatory variable, categorised into five groups: poorest, poorer, middle, richer and richest.

Statistical analysis

Data analysis was conducted using R statistical software. The survey package was used to account for the complex survey design by incorporating the person sampling weight (PERWEIGHT), primary sampling units (PSUs) and sampling strata (STRATA). PERWEIGHT was used as provided in the IPUMS DHS extract, following the IPUMS documentation. The analysis estimated weighted prevalence of adolescent childbearing within each household wealth quintile. Results are presented as percentages with corresponding 95% confidence intervals.

Results

The weighted prevalence of adolescent childbearing varied across household wealth quintiles (Table 1). Prevalence was highest among girls in the poorest household wealth quintile (37.25%, 95% CI: 27.71–46.79) and lowest among those in the richest quintile (14.76%, 95% CI: 9.56–19.97), representing an absolute difference of 22.49 percentage points. Overall, prevalence generally decreased with increasing household wealth. However, the pattern was not strictly linear, as prevalence was slightly higher in the richer quintile (27.74%) than in the middle quintile (24.84%).

Table 1. Weighted prevalence of adolescent childbearing by household wealth quintile

Household wealth quintile	Weighted prevalence (%)	95% confidence interval
Poorest	37.25	27.71–46.79
Poorer	30.05	20.44–39.65
Middle	24.84	15.45–34.23
Richer	27.74	19.20–36.27
Richest	14.76	9.56–19.97

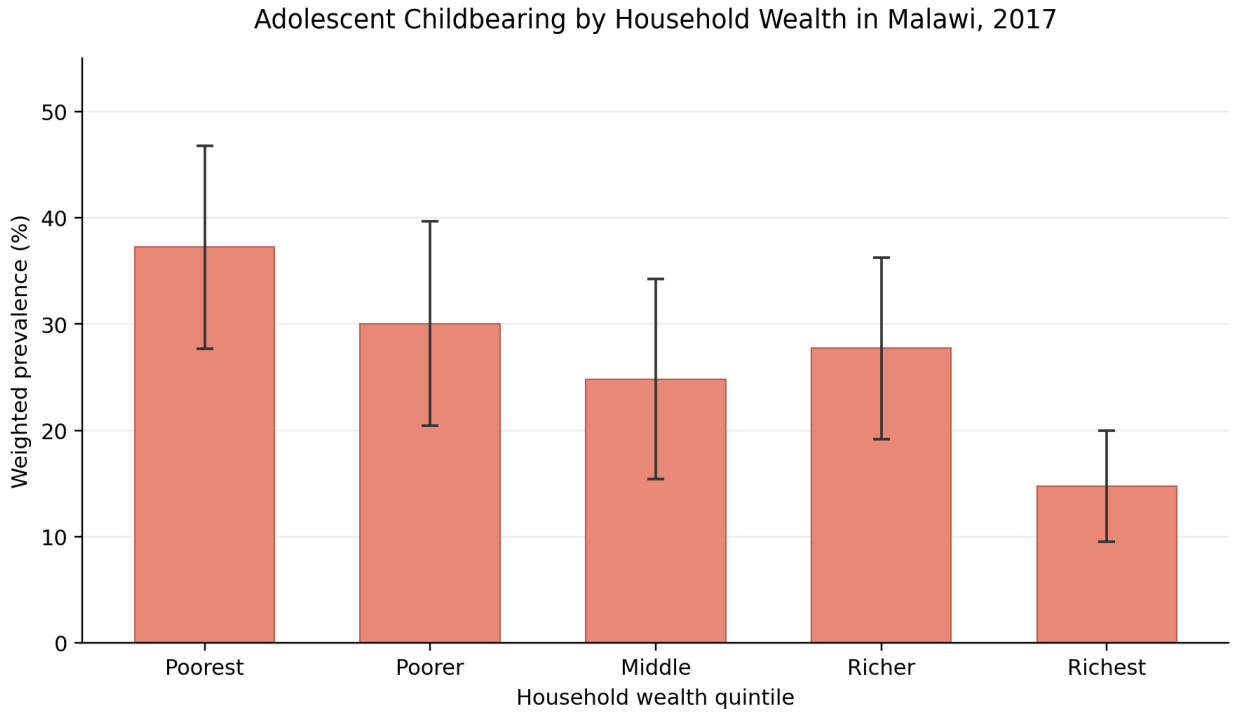


Figure 1. Weighted prevalence of adolescent childbearing among girls aged 15–19 years by household wealth quintile, Malawi DHS 2017. Error bars represent 95% confidence intervals. Analytic sample: 828 adolescents.

Discussion

This study found that adolescent childbearing varied across household wealth groups in Malawi. The weighted prevalence was 37.25% among girls in the poorest household wealth quintile, compared with 14.76% among those in the richest quintile. This 22.49 percentage-point difference highlights a substantial socioeconomic disparity in adolescent childbearing. The general decline in prevalence with increasing household wealth suggests that adolescent childbearing was more common among girls from economically disadvantaged households.

Several factors may help explain this pattern. Household economic circumstances may influence girls' educational opportunities, access to sexual and reproductive health information and services, and ability to

make reproductive decisions. Poverty may also interact with factors such as early marriage, geographical location and cultural expectations. However, these factors were not directly examined in this analysis and should be investigated in further research.

The prevalence was slightly higher in the richer quintile (27.74%) than in the middle quintile (24.84%), indicating that the relationship between household wealth and adolescent childbearing was not strictly linear. This suggests that household wealth alone may not fully explain differences in adolescent childbearing.

These findings highlight the importance of considering socioeconomic inequalities when planning adolescent sexual and reproductive health programmes in Malawi. Public health strategies could combine universal access to reproductive health information and services with additional attention to adolescents facing socioeconomic disadvantage. Potential approaches include supporting girls' continued education, strengthening youth-friendly health services and addressing economic barriers to accessing reproductive health care.

Limitations

This study has several limitations. First, the cross-sectional and descriptive nature of the analysis means that associations between household wealth and adolescent childbearing cannot be interpreted as causal relationships. Second, the data were collected in 2017 and may not reflect current patterns of adolescent childbearing in Malawi. Third, the outcome measured whether a girl had ever given birth to at least one child. It therefore does not capture pregnancies that did not result in a live birth. Finally, the analysis did not include formal statistical tests comparing prevalence across wealth quintiles. Consequently, the observed differences should be interpreted descriptively rather than as evidence of statistically significant differences.

Conclusion

Adolescent childbearing in Malawi varied across household wealth groups in the 2017 DHS, with the highest weighted prevalence observed among girls in the poorest quintile and the lowest among those in the richest quintile. These findings highlight socioeconomic inequalities in adolescent childbearing and the importance of considering household disadvantage when designing adolescent sexual and reproductive health programmes. Further research could examine how education, geographical location and other socioeconomic factors contribute to these differences.

References

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